Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Geological Society of America Bulletin

Reversal pattern and apparent polar wander for the Late Jurassic

M. B. STEINER and C. E. HELSLEY

Geological Society of America Bulletin 1975;86;1537-1543
doi: 10.1130/0016-7606(1975)86<1537:RPAAPW>2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Reversal pattern and apparent polar wander for the Late Jurassic

M. B. STEINER
C. E. HELSLEY
Institute for Geosciences, University of Texas at Dallas, Richardson, Texas 75080

ABSTRACT

A paleomagnetic study of the Upper Jurassic (Kimmeridgian-Tithonian) Morrison Formation near Norwood, Colorado, indicates the existence of thirteen polarity intervals. The seven reversed intervals occupy much more of the section than the six normal intervals, suggesting that during this time the dominant polarity was reversed. Pole positions were computed for each portion of the section where directions were tightly grouped. The pole positions form two separate groups directly related to the stratigraphic position in the section of the samples from which they were computed. The two mean pole positions for the Morrison Formation, 142.2°E, 61.4°N (dp = 4.0°, dm = 6.5°) and 161.8°E, 67.5°N (dp = 3.5°, dm = 5.0°), define a path which includes the Cretaceous pole positions for North America. The data indicate that the Jurassic apparent polar wander curve for North America is approximately a line of latitude (present-day coordinates) connecting published Triassic and Cretaceous pole positions. The data disagree with the commonly held view that the Jurassic portion of the North American apparent polar wander curve includes the present axial dipole. *Key words: solid-Earth geophysics, paleomagnetism, Jurassic, paleopole, magnetostratigraphy.*

INTRODUCTION

The position of the pole during Jurassic time has not been well defined for the North American plate. For many years, few reliable poles existed, and the various determinations diverged widely (Fig. 1). Many authors have considered the pole calculated for the White Mountain magma series (Opdyke and Wensink, 1966) to represent the Jurassic pole position, but as we have discussed previously (Steiner and Helsley, 1974a), the White Mountain pole should not be regarded as representative of the Jurassic pole.

The Upper Jurassic Summerville Formation (Steiner and Helsley, 1972) gives essentially the same pole position as do Upper Triassic units. The present paper reports the last portion of a series of studies made on Jurassic rocks (Steiner and Helsley, 1972) for the purpose of determining the Jurassic pole for North America. To determine the path of the apparent polar movement and the rate of motion, between the Late Triassic pole position and the Cretaceous position, we have undertaken a study of the Kimmeridgian-Portlandian (Kimmeridgian-Tithonian) Morrison Formation.

Although the primary purpose of the study was to derive a pole position, the sampling was designed to also provide a detailed record of the changes of polarity of the magnetic field during this time. Not only does the age of these sediments correspond to a crucial time in the North American polar wander curve, but these rocks were also being deposited at approximately the same time as the formation of the oldest portions of the magnetic anomaly sequence of the ocean floors.

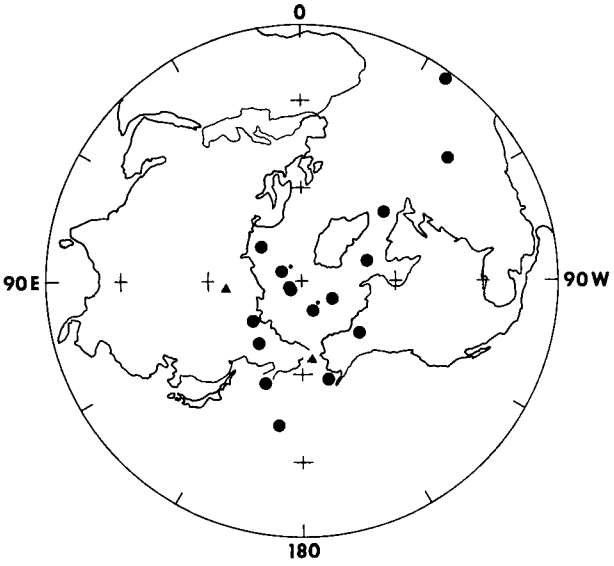


Figure 1. All published Jurassic poles for North America (solid dots). Stars mark NRM poles. Triangles indicate approximate average Triassic and Cretaceous poles.

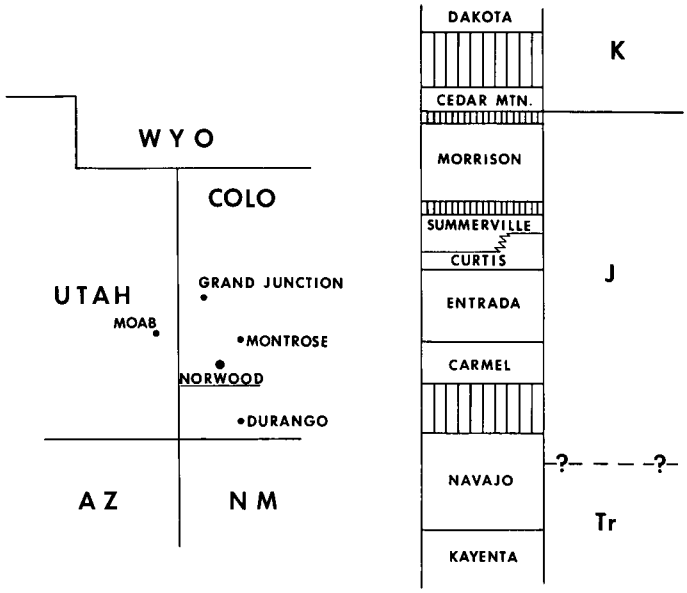


Figure 2. Location of sampling site.

Geological Society of America Bulletin, v. 86, p. 1537–1543, 8 figs., November 1975, Doc. no. 51106.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

GEOLOGY

The Morrison Formation is a fluvial-floodplain sequence of sediments deposited in a continental basin of at least 960,000 sq km (Cadigan, 1967). The formation is frequently divided into at least two units: the lower half, the Salt Wash Member, which is dominated by channel sandstones; and the upper half, the Brushy Basin Member, which is predominantly a mudstone unit. Volcanic material, present particularly in the upper mudstones, reflects tectonic activity in the western interior of the United States at that time.

The formation is a thick sequence of red, green, purple, yellow, and white sediments, commonly ranging from 150 to 250 m thick. The Morrison sediments overlie the Summerville Formation, conformably in many areas, and are often overlain by a Lower Cretaceous unit, either the Burro Canyon Formation or the equivalent age Cedar Mountain Formation. At least in some areas, the contact with these overlying units is conformable (Craig and others, 1961). Elsewhere the Morrison Formation is overlain by the Upper Cretaceous Dakota Formation.

The age of the Morrison is based primarily on its dinosaur fauna, which show close relationship to a similar African fauna of Portlandian age (Schuchert, 1918; Simpson, 1926; Stokes, 1952), and on its frequently conformable, possibly gradational (Fred Peterson, 1974, personal commun.) relationship to underlying beds containing Oxfordian ammonites (Imlay, 1952). Additional faunal evidence, summarized by Imlay (1952), suggests that the formation may be pre-Purbeckian (Late Tithonian) in age. The age of the formation, therefore, is thought to be Kimmeridigian and a portion of the Tithonian Stages.

LOCALITIES AND SAMPLING

In part of central western Colorado, the Morrison Formation is red throughout an unusually large portion of the formation. Just east of the town of Norwood (38.1°N, 108.2°W; see Fig. 2) a large roadcut (Norwood Hill) has been made for State Highway 141 in the canyon wall along the San Miguel River. This roadcut exposes the upper three-fourths of the Morrison Formation; the lowest 45 to 60 m is poorly exposed in the lower canyon walls. The lower portion of the section in the roadcut is composed predominantly of white or pink channel sandstone with interlayered red and green mudstones. The upper part is predominantly red and green mudstones

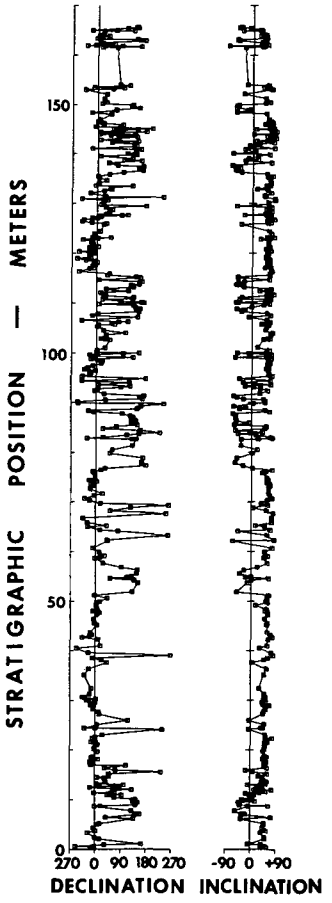


Figure 3. NRM directions of all samples in stratigraphic sequence.

with interbedded fine sandstone. The strata are flat-lying at this locality.

We drilled the well-exposed 165 m of this section at 15- to 90-cm intervals, averaging about 30 cm between samples. Samples are 2.5-cm-diam cores, oriented with a magnetic compass. At Norwood Hill, 425 samples were taken. In an attempt to avoid any large gaps in the section, the channel sandstones were sampled in detail. The procedure was to sample the host rock up to the level of the channel top off to the side of the channel and then to sample from base to top within the channel.

Near the top of the roadcut, there are about 7½ to 10½ m of greenish mudstone and sandstone with 4.5 m of red mudstone above. Overlying this sequence are conglomerate, conglomeratic sandstone, and interbedded mudstone; these are believed to be part of the Lower Cretaceous Burro Canyon Formation (E. M. Shoemaker, 1975, personal commun.; Bush and others, 1959). Above these, the sandstone and interbedded coals of the Dakota Formation rest. The coals contain lower Albian palynomorphs (J. B. Urban, 1974, personal commun.).

TREATMENT

In the laboratory, samples were cut into 2.5-cm-long cylinders. NRM measurements were made on a PAR SM-1 spinner magnetometer. Samples were thermally demagnetized in a horizontal, noninductively wound furnace of 18 sample capacity and cooled in two nested mu-metal cans which reduce the field to less than 10 gammas. After the first demagnetization step, samples were measured on the PAR magnetometer and, after higher demagnetization steps, on the ScT cryogenic magnetometer. All equipment was housed in a 3-m cube mu-metal room; thus samples were not exposed to fields greater than 400 γ, once demagnetization was begun. Sample directions were plotted on both orthogonal axes projections and on Schmidt equal-area projections as demagnetization proceeded. The temperature considered to give the best direction for a particular sample was judged by a combined assessment of the demagnetization paths on both plots. That is, a persistent erratic behavior on one or the other or both types of plots indicates that the best demagnetization temperature has been exceeded. The direction obtained with the temperature step prior to this behavior was considered the best direction for the sample.

DATA

The NRM directions of the Norwood Hill samples (Fig. 3) indicate the presence of both polarities in an alternating sequence. There are groups of sequential samples in most of the polarity zones which have well-clustered directions. However, a large portion of the sample directions cluster around the present geocentric axial dipole direction or lie in a streak between the axial dipole direction and the typical Morrison directions. (The secondary magnetization direction seems to be that of the Cenozoic axial dipole field direction rather than the present field direction.) The intensity of the samples, averaging about 3.5×10^{-6} emu/cc, is somewhat lower than other continental red-bed sequences which often have NRM intensities of 5×10^{-6} emu/cc or higher.

A group of pilot samples was stepwise demagnetized in 50° to 100°C increments from 150° to about 650°C. The behavior of the samples was extremely varied, perhaps reflecting the variations of lithology exhibited by the difference in color, grain size,

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

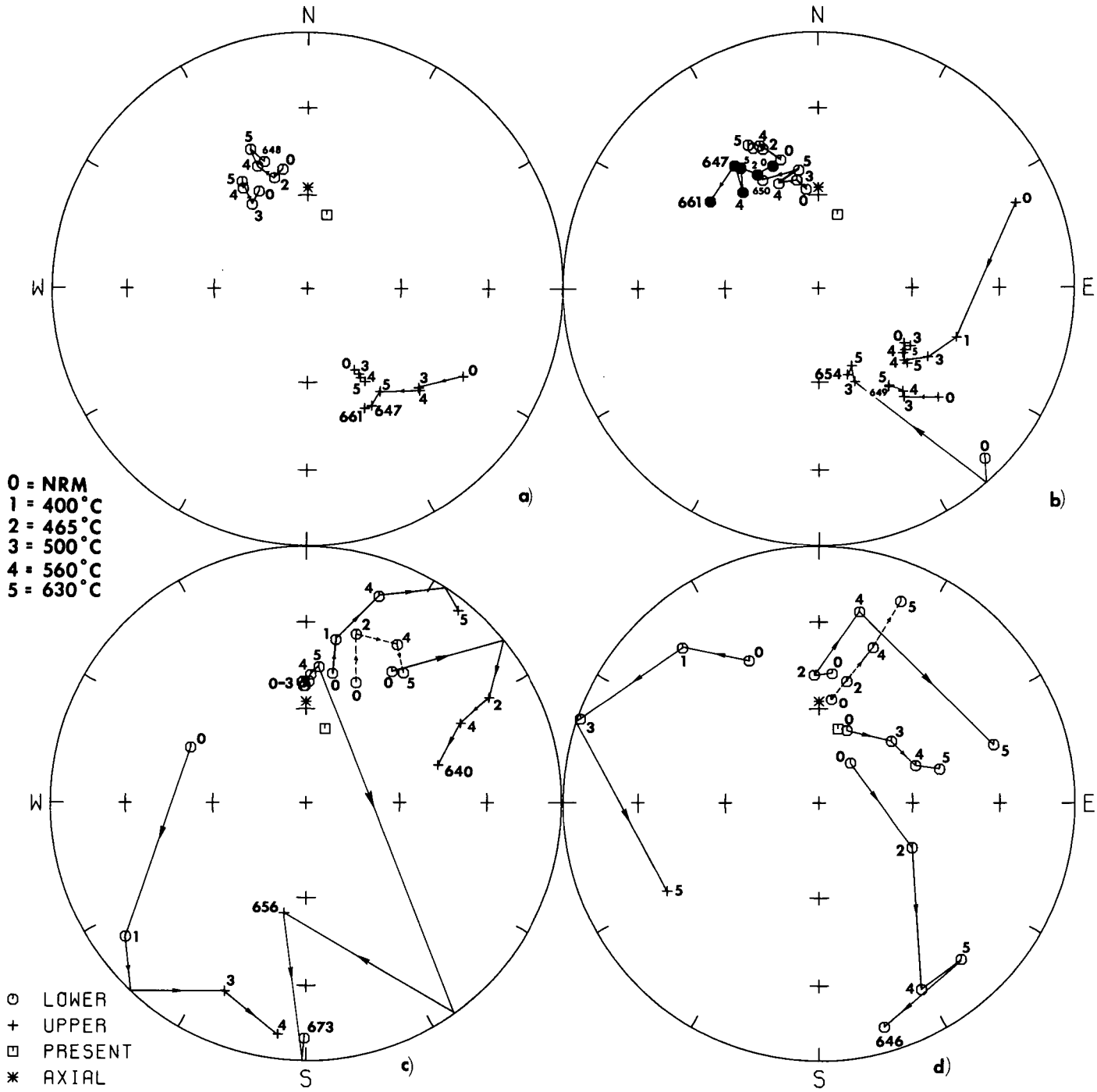


Figure 4. Typical thermal demagnetization paths. 4a and 4b represent sample behavior within well-grouped intervals. 4c and 4d show types of behavior considered indicative of reversed polarity in less well defined intervals of polarity. Temperatures are given by the key in the center bottom of the figure. Temperatures above 630°C are indicated directly on the diagrams.

and degree of cementation. Numerous further pilot samples were run until some pattern was obtained by which to predict the behavior of the rest of the samples. Figure 4 shows some typical demagnetization paths. In general, samples with lower NRM intensities (0.5×10^{-6} emu/cc) often reached their best demagnetized direction at lower temperatures (400° to 500°C) than stronger samples (5.0×10^{-6} emu/cc) and thereafter, on higher heating, diverged increasingly from the mean of the group. By that time, intensities were sufficiently low that laboratory IRM was becoming significant.

Most samples other than those of low NRM intensity were heated to 450°, 560°, and 630°C. Approximately three-fourths of all samples had to be heated to between

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

620°C and 630°C before directions began to stabilize. The NRM directions of the groups of stratigraphically consecutive samples with well-clustered directions had the least secondary moment. These samples were also heated above the 630°C step, to temperatures of 640°C to 660°C. The directions of many of these continued changing slightly at some of the higher temperatures. Because of the large number of samples in the study, only these samples were subjected to the higher heatings. In all, each sample of the total 425 was heated at least 3 to 5 times.

Even after extensive demagnetization, the directions of many of the samples appear to retain a large component of the axial field direction. Figure 5 is a stratigraphic presentation of the declinations and inclinations obtained after optimum thermal demagnetization. Thirteen polarity zones are present, which have been given numbers from the base upward for convenience of discussion. The intervals which had well-grouped NRM directions remain well grouped after demagnetization. These are preceded and succeeded by intervals in which directions are more dispersed. These latter directions are generally consistent with the idea of a significant addition of axial field component to either a normal or reversed direction. The assignment of polarity is based primarily on the demagnetization behavior of the samples. As the axial field direction is close to the normal direction for these rocks, there is a difficulty in determining

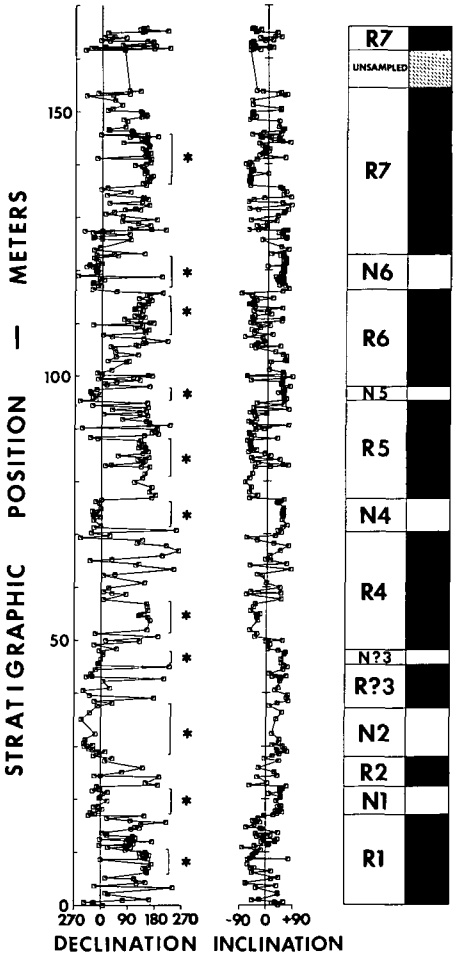


Figure 5. Directions of thermally demagnetized samples. Stars and brackets indicate the well-grouped intervals of directions from which poles were calculated.

whether a sample possesses normal polarity with a high stability overprint of the axial field direction or whether the sample direction was reversed and has been completely remagnetized in the axial field direction. Therefore, if even a few sample directions within an interval of poorly defined polarity show a tendency to move along a great circle in the direction of reversed polarity (as shown in Figs. 4c and 4d), the interval is considered to be probably reversed.

Using these criteria, we obtain the polarity column shown on the right of Figure 5. The sequence is dominated in relative duration by reversed polarity, but the number of intervals of normal and reversed polarity are about the same. The difference in thickness of the normal and reversed intervals is not a result of changes in sedimentation rate. The reversal sequence from the upper half of the formation has been observed to be the same at another locality approximately 80 km to the north (Steiner and Helsley, 1975). In addition, the Morrison reversal sequence correlates well with the reversal sequence of about the same age, derived from the oldest observed magnetic anomalies in the sea floor (Steiner and Helsley, 1975). The same relatively greater amount of reversed polarity is also obvious in the anomaly sequence.

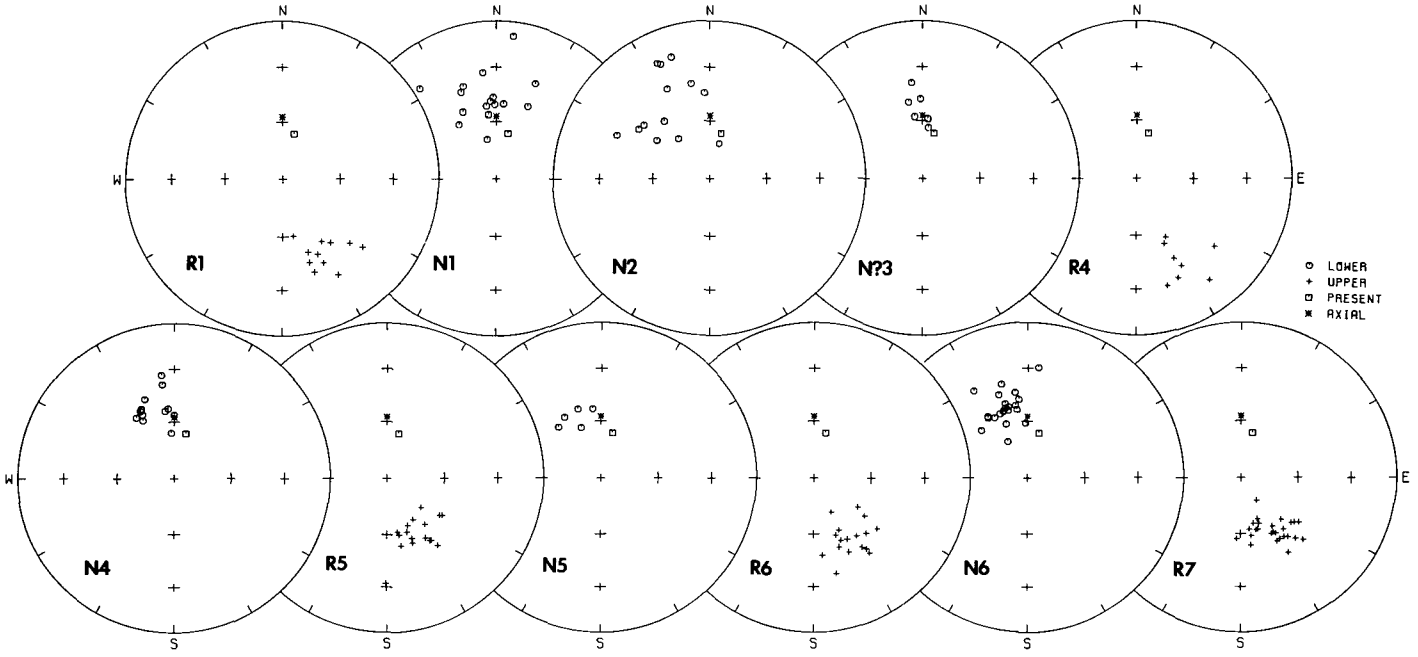


Figure 6. Directions of well-grouped intervals within each polarity zone.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 1. MEAN DIRECTIONS AND POLE POSITIONS FOR THE MORRISON FORMATION

	Declinations	Inclinations	N	R	K	α_{95}	Long. (E)	Lat (N)	dp	dm
Unit weight to intervals										
Lower unselected	155.0	−44.4	5	4.97	153.17	6.2	139.5	65.7	4.9	7.8
Upper unselected	154.3	−53.6	5	5.00	399.04	3.8	158.9	69.0	3.7	5.4
Lower selected	150.3	−42.4	3	3.00	549.42	5.3	142.2	61.4	4.0	6.5
Upper selected	152.3	−54.2	4	4.00	662.96	3.6	161.8	67.5	3.5	5.0
Unit weight to samples										
Lower unselected	154.7	−45.0	51	48.99	24.92	4.1	140.8	65.8	3.3	5.2
Upper unselected	154.9	−54.1	90	87.55	36.38	2.5	159.7	69.5	2.5	3.5
Lower selected	150.1	−42.8	32	30.66	23.23	5.4	143.0	61.4	4.1	6.7
Upper selected	152.4	−55.0	68	66.09	35.08	3.0	163.8	67.8	3.0	4.2
R1	150.8	−44.3	11	10.78	45.10	6.9	144.2	62.6	5.4	8.6
R4	151.3	−38.6	8	7.79	32.85	9.8	136.5	60.4	6.9	11.6
N1	341.2	43.3	7	6.77	25.69	12.1	128.4	69.5	9.4	15.1
N2	328.6	44.3	13	12.13	13.83	11.6	146.5	60.9	9.1	14.5
N3	357.0	53.1	6	5.93	69.68	8.1	(102.0	85.0)	7.8	11.2
N4	344.2	50.8	12	11.79	53.25	6.0	139.4	75.5	5.4	8.1
R5	151.9	−55.8	17	16.63	43.68	5.5	166.2	67.6	5.6	7.8
R6	149.0	−51.6	17	16.65	46.19	5.3	158.2	64.1	4.9	7.2
R7	154.7	−57.2	28	26.98	26.54	5.4	168.8	70.1	5.7	7.9
N5	333.8	52.1	6	5.90	48.48	9.7	155.6	68.1	9.1	13.3
N6	341.9	50.9	22	21.59	50.95	4.4	143.5	73.8	4.0	5.9

Selected = Intervals R1, R4, N2, and R5, R6, R7, N5 only.

The groups of clustered directions within the polarity zones (Fig. 5) constitute the most antipodal normal and reversed directions after demagnetization. Within some reversed groups, a few samples show directions intermediate between the reversed direction and the axial field direction. These samples were not used in the calculation of pole positions. The directions of the well-grouped normal and reversed samples without the intermediate samples are shown in Figure 6. Even within these groups, most of the normal and reversed mean directions (Table 1) are not entirely antipodal. The difference, about 10°, may be due to incomplete removal of an axial field component imprinted on both normal and reversed directions. It is peculiar that, in the well-grouped intervals, most of the sample directions in the reversed groups cluster well, but in some of the normal groups, directions are more dispersed (Fig. 6). Another peculiarity of the normal groups is the failure of individual sample directions to move very much or very consistently away from the axial field upon demagnetization. The difference in behavior cannot be identified with any obvious lithologic characteristic.

Considering the variety of rock types sampled in the Morrison Formation, it is interesting to observe which types carry the more stable remanences. Two significantly different lithologic types, very pale red channel sandstone and purplish-gray and grayish-red well-cemented mudstones showed the best grouped normal and reversed directions with the least addition of a secondary component. Upon demagnetization, these groups remained well clustered. The significant difference in color and in grain size between these stable groups and the very different stability of other rock units outwardly identical to these underscores the fact that physical appearance of sedimentary rocks is not necessarily a reliable indicator of suitability for paleomagnetic studies.

Another group of rocks which carried stable remanence were white channel sandstones, occurring at two localities in the Morrison Formation. These white sandstones are fine to medium grained; in the lower portions of the channels, they enclose laminae of black heavy minerals. The immediately overlying layers in the channel have disseminated heavy minerals in amounts varying from about 10 percent to less than 1 percent. The beds with heavy mineral laminae show a water-washed pattern on exposed laminae surfaces, clearly indicating that the black mineral had been detritally deposited. The sample directions show little change in direction when heated to temperatures in excess of 600°C, indicating that the grains are now hematite. The sample directions replicate very accurately the cleaned either normal or reversed direction observed in the rest of the formation. Samples of those higher sands with disseminated black grains duplicate the cleaned direction and show stability in degrees, apparently related to the amount of black grains contained. These preliminary observations suggest that detrital remanence may be significant in portions of the Morrison Formation.

Intensities of the Morrison samples generally dropped 30 percent upon thermal demagnetization, from an average 3.5×10^{-6} emu/cc NRM intensity to an average of 1.1×10^{-6} emu/cc. However, the majority of the samples are in the range 0.04×10^{-6} emu/cc to 1.5×10^{-6} emu/cc after demagnetization (0.6×10^{-6} to 8.0×10^{-6} emu/cc NRM). The intensity of these rocks after demagnetization is considerably lower than that of many of the Triassic red beds studied paleomagnetically (Picard, 1964; Helsley, 1969) but is comparable to that of the stratigraphically underlying Summerville Formation (Steiner and Helsley, 1972). Whether the low intensity reflects the peculiar lithology of the Morrison, and therefore environment of deposition, or whether the Morrison and Summerville together reflect something about the Jurassic geomagnetic field is difficult to say. It is interesting to speculate that the low intensity of these rocks could reflect a low-intensity Jurassic field and thus help to explain the chronic instability of many Jurassic rocks world-wide.

POLE POSITION

Poles were computed from each interval defined by a directional cluster (Fig. 5). The poles form two groups directly related to the stratigraphic position of the intervals from which the samples were taken (Fig. 7). Poles from the lower portion of the strati-

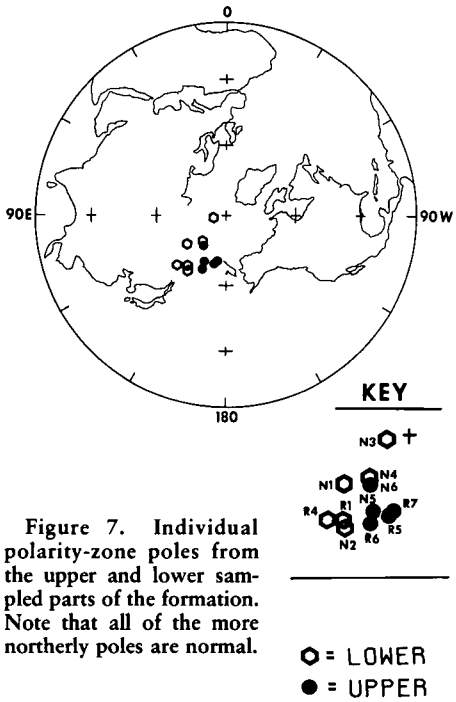


Figure 7. Individual polarity-zone poles from the upper and lower sampled parts of the formation. Note that all of the more northerly poles are normal.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

graphic sequence have noticeably smaller values of east longitude than poles from the upper portion. Poles calculated for the reversed intervals in particular cluster into these two groups. The lower and upper normal poles seem to lie in a streak between their respective associated reversed poles and the present axial dipole. (Both N3 and the upper half of N2 have directions very similar to the axial dipole field direction. These were not used in the calculation of mean poles.)

Since the poles naturally cluster into two groups, two mean poles were computed for the Morrison Formation. The difference in the poles is interpreted as reflecting apparent polar motion during Morrison deposition. The two mean poles have been computed in two ways. In both the upper and the lower portion of the section, the reversed poles and the one normal pole most removed from the axial dipole form very tight groups (Fig. 7). Thus, the two mean poles were computed from both, all poles in each part of the section ("Unselected," Table 1), and from only the reversed poles and the normal pole most removed from the axial dipole ("Selected"). Further, the mean poles were computed by (1) giving unit weight to the individual poles and (2) giving unit weight to all samples used in computation of the individual poles (Table 1).

Since one normal pole occurs within each group of reversed poles, whereas the rest string out toward the axial dipole, it is felt that the direction of the majority of the normal samples must still contain an appreciable secondary moment. The alternate explanation that the normal and reverse fields were different has no supporting evidence at this time. The poles within the "selected" groups probably have no secondary component remaining, because, if there were any, removal of it from both normal and reversed directions could only cause these tight groups to become more dispersed. Thus, the two means computed from these poles probably are the most reliable mean poles. The mean poles computed by giving unit weight either to polarity-zone group means or to individual samples give approximately the same pole positions and ovals of confidence. Either is probably an accurate estimate of the pole position. The means of the interval poles are used for comparisons in this paper.

COMPARISON WITH PUBLISHED DATA

The data acquired from the Morrison Formation indicate an apparent motion of the pole relative to North America from the vicinity of the underlying Summerville Formation pole at 86.3°E, 70.1°N (5.1, 8.7) (Steiner and Helsley, 1972) to 142.2°E, 61.4°N (4.0, 6.5) by early Morrison time and to 161.8°E, 67.5°N (3.5, 5.0) by late Morrison time (Fig. 8). These data confirm our earlier speculations that the path of Jurassic apparent polar motion for North America is approximately along a line of present-day latitude, and that poles lying in the vicinity of the geocentric axial dipole, such as the White Mountain magma series, are not definitive evidence that the North American polar wander path passed through the axial pole position during Jurassic time. The poles obtained from the Topley Intrusions (Symons, 1973) support the path indicated by the Morrison poles, although the dispersion among the Topley site mean poles is quite high. DeBoer (1967) also obtained a pole in a position very similar to the lower Morrison pole from dikes dated only as cutting Upper Triassic sediments. The Morrison pole supports DeBoer's speculation that the dikes may be Jurassic. The majority of published Cretaceous poles from North America lie closer to 180° long. than the upper Morrison poles, indicating that the apparent movement of the pole was not entirely completed by the end of magnetization of the Morrison Formation sampled here.

The movement of the pole through Late Jurassic time reflects the early opening of the Atlantic. The Morrison results place certain constraints upon the initial motions of the North American continent that are not entirely in agreement with those derived from sea-floor data (Steiner, 1975).

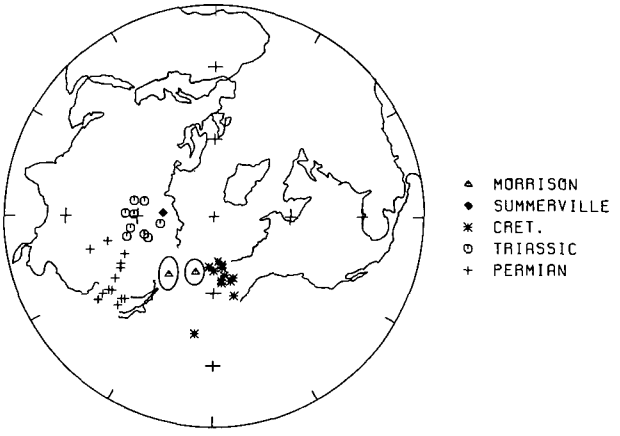


Figure 8. The two mean Morrison poles computed from the means of the reversed intervals and those normal intervals most removed from the axial dipole are plotted with their ovals of confidence. Permian, Triassic, and Cretaceous poles as summarized by McElhinny (1973) are also plotted. (Triassic poles have been selected as in Steiner, 1975, with poles from Reeve and Helsley, 1972, Steiner and Helsley, 1974b, and Helsley and Steiner, 1974, added.) Cretaceous poles also include two unpublished poles by Helsley and that of Shive and Frerichs (1974).

DISCUSSION AND CONCLUSION

The data from the Morrison Formation at Norwood, Colorado, indicate that the Late Jurassic was a time of frequent reversals of the magnetic field with reversed polarity predominating. Thirteen intervals of polarity can be defined. The dominance of reversed polarity is seemingly not accompanied by a decrease in the number of times the field changed to normal polarity. This combination may have implications for the field reversal mechanism.

Two distinct groups of poles were obtained from the lower and upper parts of the formation. The two groups are sequential in position and time on a path leading to the average Cretaceous pole position, but the individual poles within the two groups do not present a continuum of pole positions. The two groups may be evidence that the upper and lower parts of the formation were magnetized at two slightly different times with a time break in between, although the field evidence does not favor such an explanation. There actually may have been continuous magnetization (and/or deposition), but because of the small portion of the magnetic record that was preserved well enough to allow reliable pole calculations, a fortuitous two-group pattern is observed. It is also possible that the apparent motion of the pole was discontinous, perhaps caused by a discontinuity in plate motions.

Regardless of the reason, the two poles define two points on a path between the Triassic and Cretaceous poles for North America. The data, therefore, present strong evidence that the Mesozoic apparent polar wander curve for North America is approximately along a line of present-day latitude between the known Triassic and Cretaceous pole positions. The path described by these data disagree with the commonly held view that the Jurassic portion of the apparent polar wander curve includes the present axial dipole.

ACKNOWLEDGMENTS

We gratefully acknowledge the assistance of Joan and Ann Helsley in the collection of the samples. Norma Booth provided valuable assistance in the measurement of

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

some of the samples. We thank Beverly McMahon for critically reading the manuscript. This study was supported under National Science Foundation Grant GA–15999.

REFERENCES CITED

Bush, A. L., Bromfield, C. S., and Pierson, C. T., 1959, Area L geology of the Placerville quadrangle, San Miguel County, Colorado: U.S. Geol. Survey Bull. 1072-E, p. 330–333.

Cadigan, R. A., 1967, Petrology of the Morrison Formation in the Colorado Plateau: U.S. Geol. Survey Prof. Paper 556.

Craig, L. C., Ekren, E. B., Houser, F. N., Shawe, D. R., Simmons, G. C., and Katich, P. J., Jr., 1961, Dakota Group of the Colorado Plateau: Discussion: Am. Assoc. Petroleum Geologists Bull., v. 45, p. 1582–1592.

DeBoer, J., 1967, Paleomagnetic-tectonic study of Mesozoic dike swarms in the Appalachians: Jour. Geophys. Research, v. 72, p. 2237–2250.

Helsley, C. E., 1969, Magnetic reversal stratigraphy of the Lower Triassic Moenkopi Formation of western Colorado: Geol. Soc. America Bull., v. 80, p. 2431–2450.

Helsley, C. E., and Steiner, M., 1974, Paleomagnetism of the Lower Triassic Moenkopi Formation: Geol. Soc. America Bull., v. 85, p. 457–464.

Imlay, R. W., 1952, Correlation of the Jurassic formations of North America exclusive of Canada: Geol. Soc. America Bull., v. 63, p. 953–992.

McElhinny, M. W., 1973, Paleomagnetism and plate tectonics: Cambridge, England, Cambridge Univ. Press, p. 285–286.

Opdyke, N. D., and Wensink, H., 1966, Paleomagnetism of rocks from the White Mountain plutonic-volcanic series in New Hampshire and Vermont: Jour. Geophys. Research, v. 71, p. 3045–3051.

Picard, M. D., 1964, Paleomagnetic correlation of units within Chugwater (Triassic) Formation, west-central Wyoming: Am. Assoc. Petroleum Geologists Bull., v. 48, p. 269–291.

Reeve, S. C., and Helsley, C. E., 1972, Magnetic reversal sequence in the upper portion of the Chinle Formation, Montoya, New Mexico: Geol. Soc. America Bull., v. 83, p. 3795–3812.

Schuchert, C., 1918, Age of the American Morrison Formation and East African Tendaguru Formations: Geol. Soc. America Bull., v. 29, p. 245–289.

Shive, P. N., and Frerichs, W. E., 1974, Paleomagnetism of the Niobrara Formation in Wyoming, Colorado, and Kansas: Jour. Geophys. Research, v. 79, p. 3001–3008.

Simpson, G. G., 1926, The age of the Morrison Formation: Am. Jour. Sci., 5th ser., v. 12, p. 198–216.

Steiner, M., 1975, Mesozoic apparent polar wander and Atlantic plate tectonics: Nature, v. 254, p. 107–109.

Steiner, M., and Helsley, C. E., 1972, Jurassic polar movement relative to North America: Jour. Geophys. Research, v. 77, p. 4981–4993.

Steiner, M., and Helsley, C. E., 1974a, Paleomagnetic results from the Jurassic Topley Intrusions near Endako, British Columbia: Discussion: Canadian Jour. Earth Sci., v. 11, p. 349–350.

——1974b, Magnetic polarity sequence of the Upper Triassic Kayenta Formation: Geology, v. 2, p. 191–194.

——1975, The Late Jurassic magnetic polarity sequence: Earth and Planetary Sci. Letters (in press).

Stokes, W. L., 1952, The Lower Cretaceous in the Colorado Plateau: Am. Assoc. Petroleum Geologists Bull., v. 36, p. 1766–1776.

Symons, D.T.A., 1973, Paleomagnetic results from the Jurassic Topley Intrusions near Endako, British Columbia: Canadian Jour. Earth Sci., v. 10, p. 1099–1108.

MANUSCRIPT RECEIVED BY THE SOCIETY SEPTEMBER 20, 1974
REVISED MANUSCRIPT RECEIVED MARCH 24, 1975
MANUSCRIPT ACCEPTED APRIL 4, 1975
CONTRIBUTION NO. 265, INSTITUTE FOR GEOSCIENCES, THE UNIVERSITY OF TEXAS AT DALLAS, RICHARDSON, TEXAS

Printed in U.S.A.